CHAPTER 4

FIELD EFFECT TRANSISTORS (FETs)

FETs are :

- three terminal devices, these terminals are Drain, source and gate
- Unipolar device.
- current through it is due to flow of only one of the two kinds of charge carriers.(holes or electrons)
- The name field effect is derived from the fact that the current flow is controlled by an electric field; set up in the device by an externally applied voltage.
- FETs are a voltage controlled device whereas BJTs are current controlled device.
- It has high input impedance.

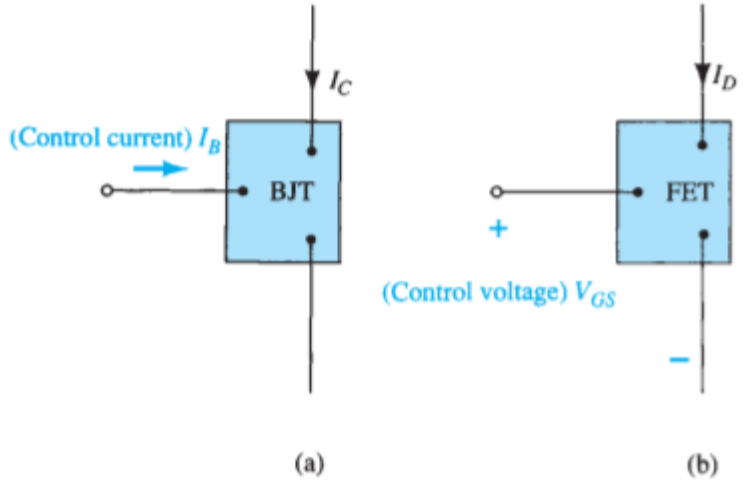
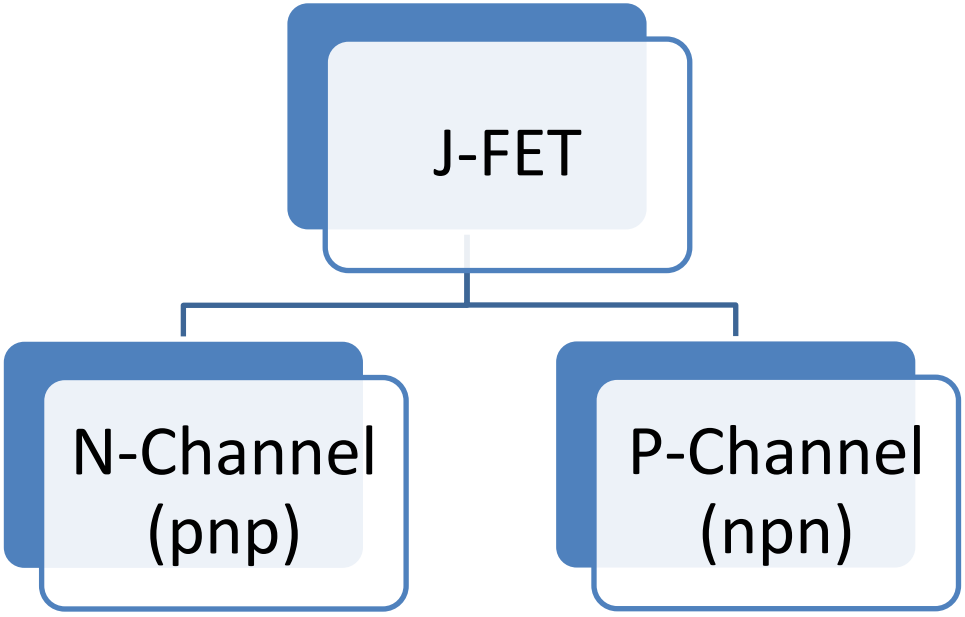


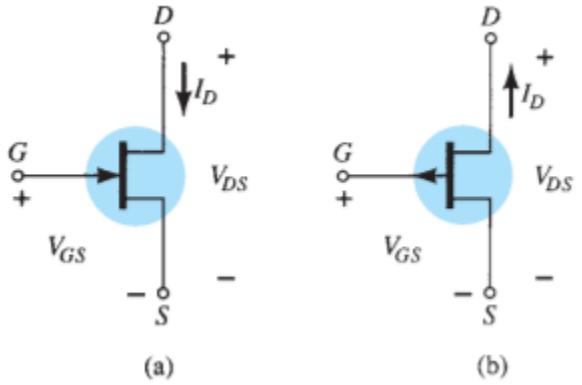
Fig. (a) Current-controlled and (b) voltage-controlled amplifiers.

- Voltage gains for BJT amplifiers are great deal more than for FETs.
- FETs are more temperature stable than BJTs.
- FETs are smaller in size than BJTs. (useful for IC chips construction)

There are two main types of FETs:-

1) Junction field effect transistor (J-FET)
2) Metal oxide semiconductor field effect transistor (MOSFET)





JFET symbols: (a) n-channel; (b) p-channel

Construction and characteristics of JFETs

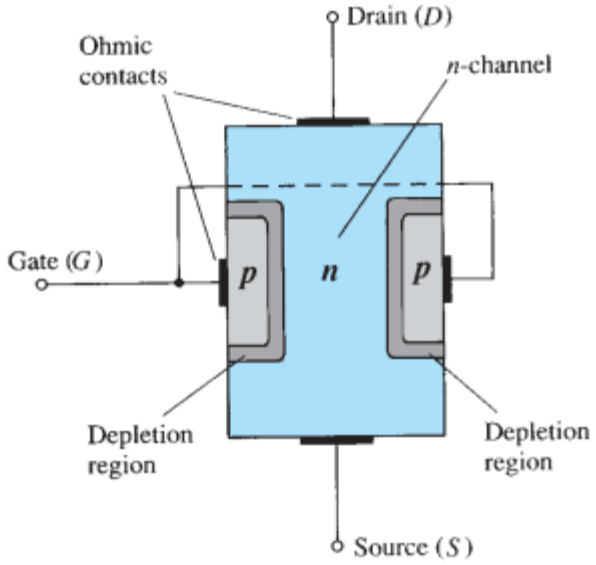
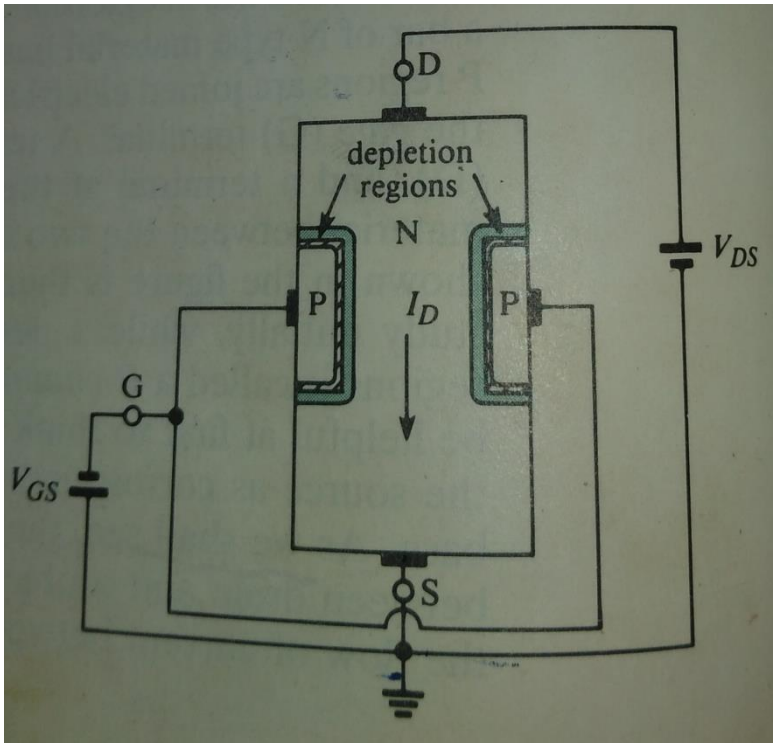


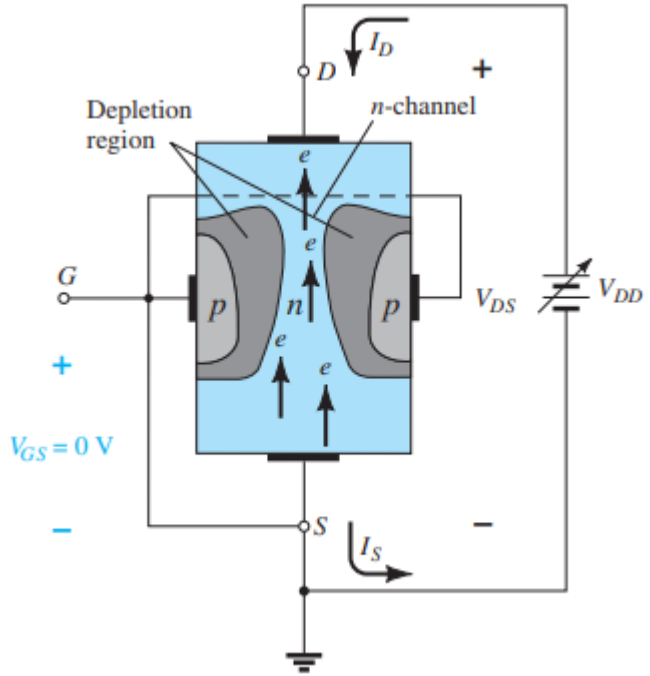
Fig. N-channel J-FET

- ✓ **The major part of the structure is the N-type material, which forms the channel between the embedded layers of p-type material.**
- ✓ **The two p-type materials are connected together to the gate terminals.**

N-channel J-FET operation



- ✓ **When an external voltage is connected between the drain and source of the N-channel J-FET.**
- ✓ **Drain is positive with respect to the source, current is established by the flow of electrons through the N-material from the source to the drain.**



Source- origin of electrons.

$$I_D = I_S$$

- ✓ **Width of the depletion region depends of the value of V_{GS} - reverse bias voltage.**
- ✓ **As V_{GS} is more negative, depletion expand and width of the channel decreases.**
- ✓ **↓channel = ↑resistance of the channel**

  =↓flow of current I_D.
- ✓ **PN-junction on each side of the channel are reverse biased, due to voltage applied across gate and source. $I_G = 0$.**
- ✓ **The channel is more lightly doped than the gate, so the depletion region penetrates more deeply to the N-type channel than into p-type material.**
- ✓ **For $V_{GS} = 0V$ $(conistant)$, As V_{DS} increases I_D $increases$.**

✓ **The maximum current occurs when $V_{GS} = 0V$.**

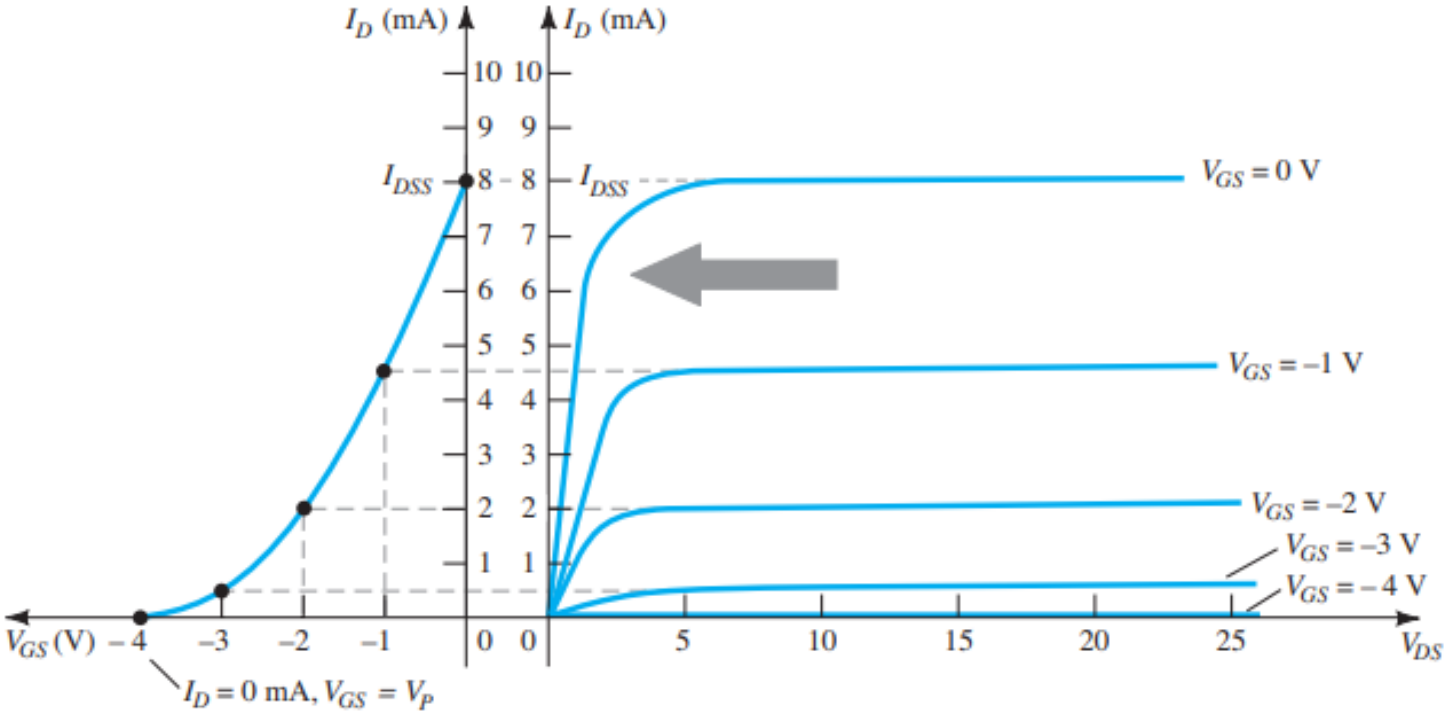
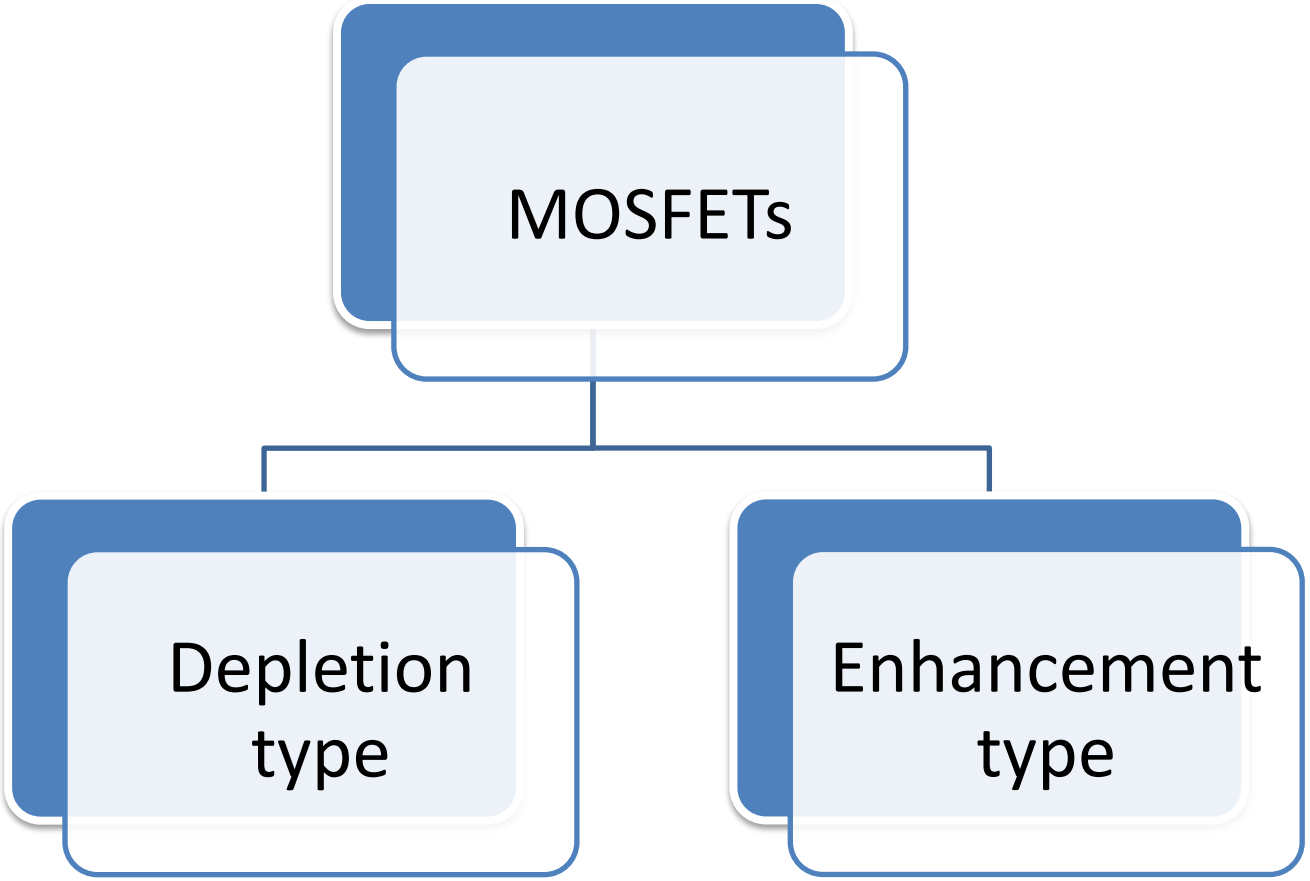
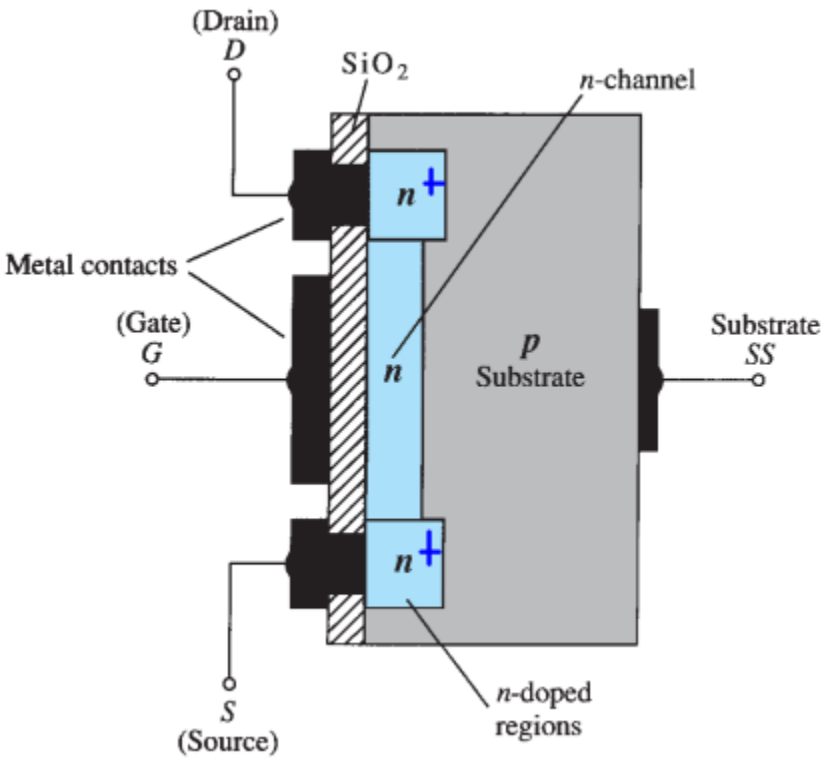


Fig. VI-characteristic curve for N-channel J-FET


MOSFETs
Depletion type
Enhancement type

- ✓ **MOSFET is similar to JFET in many things. Three terminals (D,S,G), voltage controlled devices (V_{GS}).**
- ✓ **Their difference is the gate terminal in MOSFET is insulated from its channel region.**
- ✓ **For this reason, MOSFET is said to be insulated gate field effect transistor. (IGFET)**

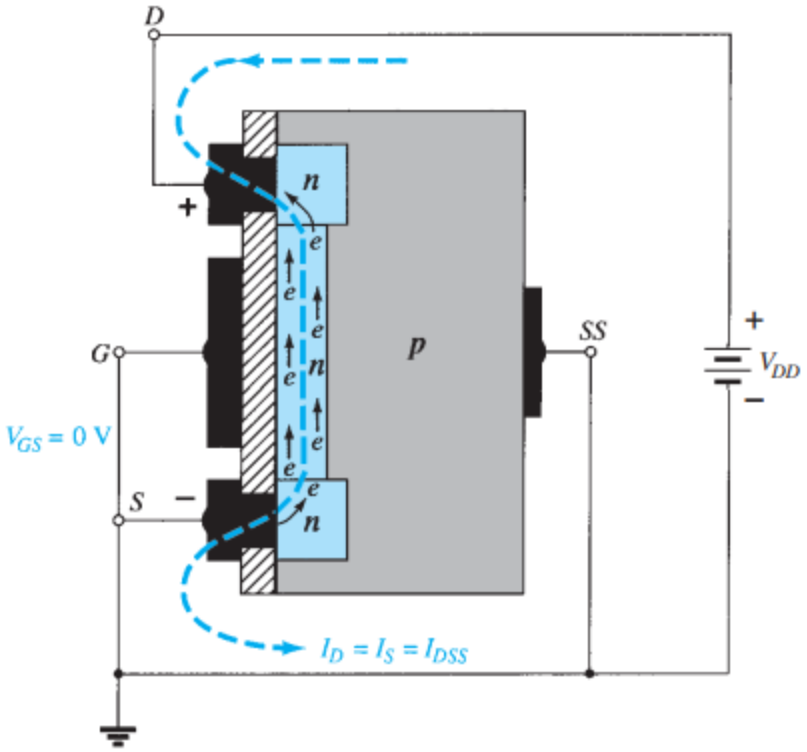
Depletion type MOSFETs (N-channel)



- ✓ **A p-type silicon forms a substrate, in which there are embedded two heavily doped and one lightly doped N-type silicon.**
- ✓ **A thin layer of SiO_2, is deposited along the surface.**
- ✓ **Metal contacts penetrate the SiO_2 layer at the two N^+ wells at drain and source terminals.**

- ✓ **Between the two N^+ wells there is a more lightly doped region of N material that forms the channel.**
- ✓ **Metal (Al) is deposited on the SiO_2, which is in opposite side of the channel and becomes the gate terminal.**
- ✓ **SiO_2 isolates the gate from channel.**
- ✓ **Insulating layer of SiO_2 in the MOSFET construction creates high input impedance of the device.**
- ✓ **Results in $I_G = 0A$ for dc-biased configuration.**

Basic operation and characteristics for D-MOSFET



- ✓ A voltage V_{DD} is connected between drain and source to make the drain positive with respect to the source.
- ✓ Substrate is always connected to the source.
- ✓ When the gate is made negative with respect to the source by V_{GS}, the electric field it produces in the channel drives electrons away from the portion of the channel near the SiO_2 layer.
- ✓ This portion is thus depleted of free carriers and the channel width is narrowed.

- ✓ **The narrower the channel, the greater the resistance and the smaller the current flow from drain to source.**

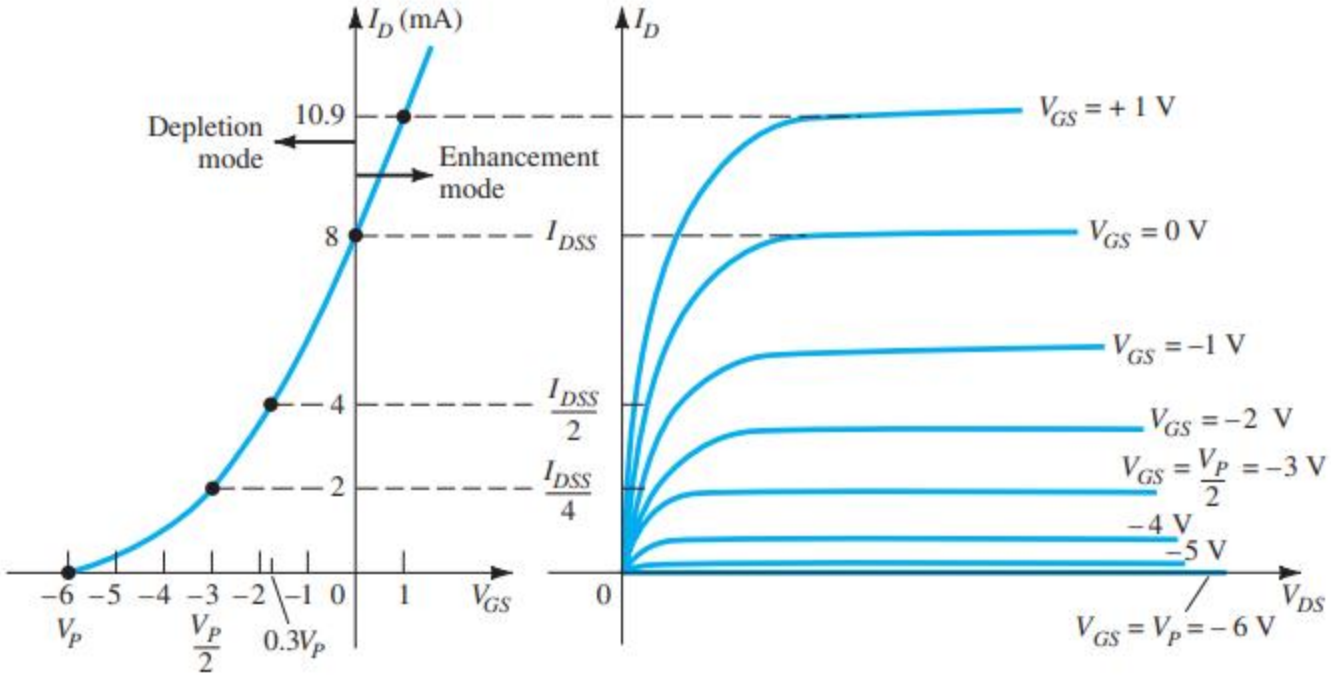
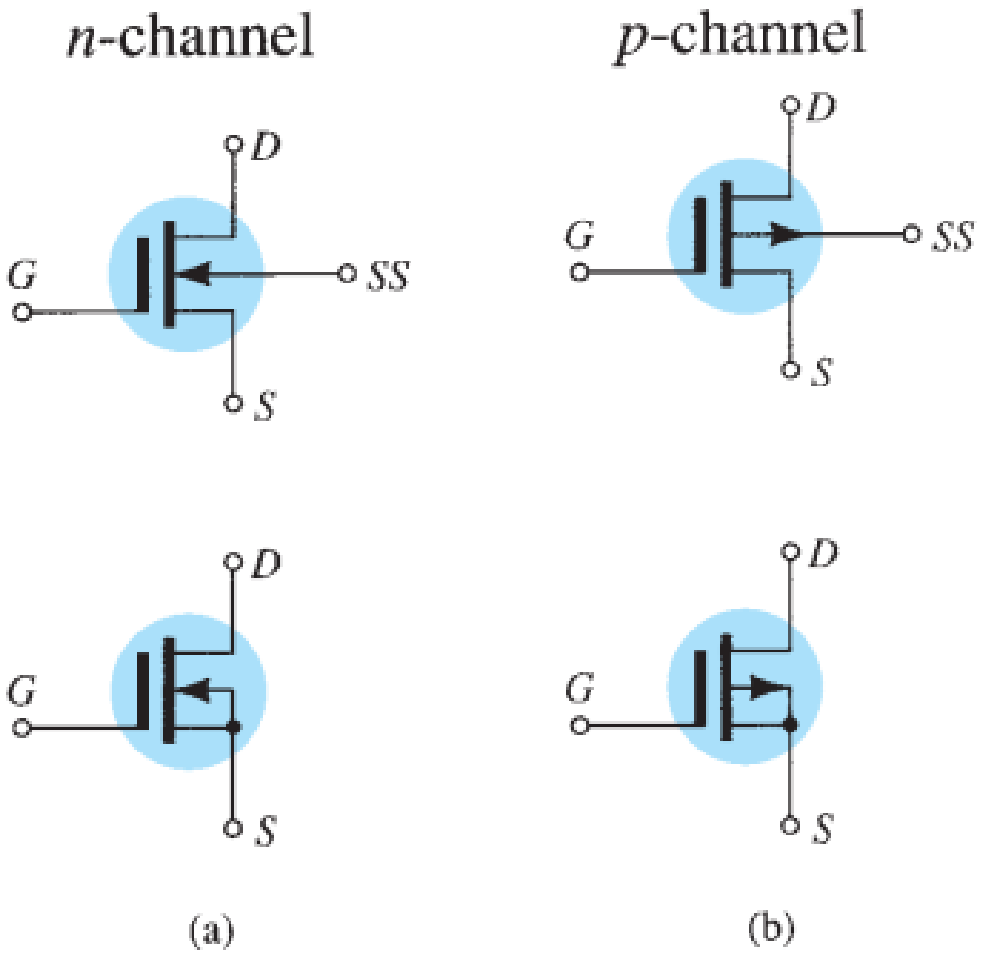
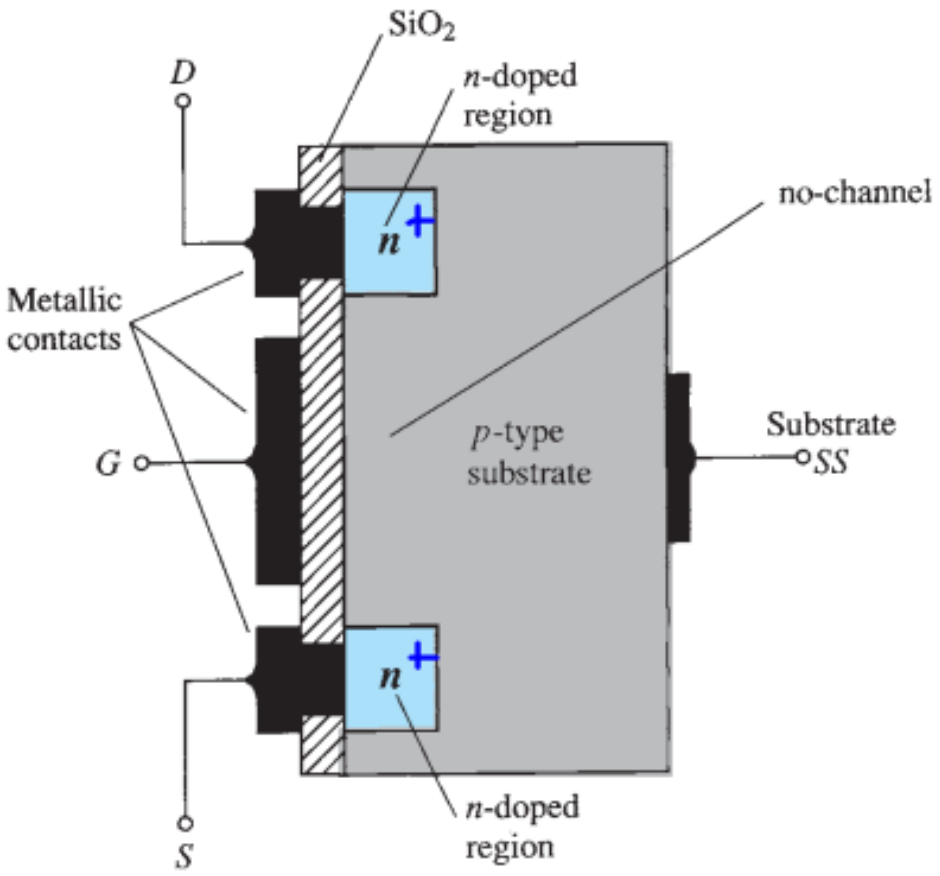


Fig. VI-characteristic curve for N-channel D-MOSFET



Graphic symbols for:
(a) n-channel depletion-type MOSFETs and (b) p-channel depletion-type MOSFETs.

N-channel Enhancement type-MOSFET



- ✓ **In the Enhancement MOSFET, there is no N-type material between the drain and source; instead the p-type substrate extends all the way to SiO_2 layer adjacent to the gate.**

Basic operation and characteristics for E-MOSFET

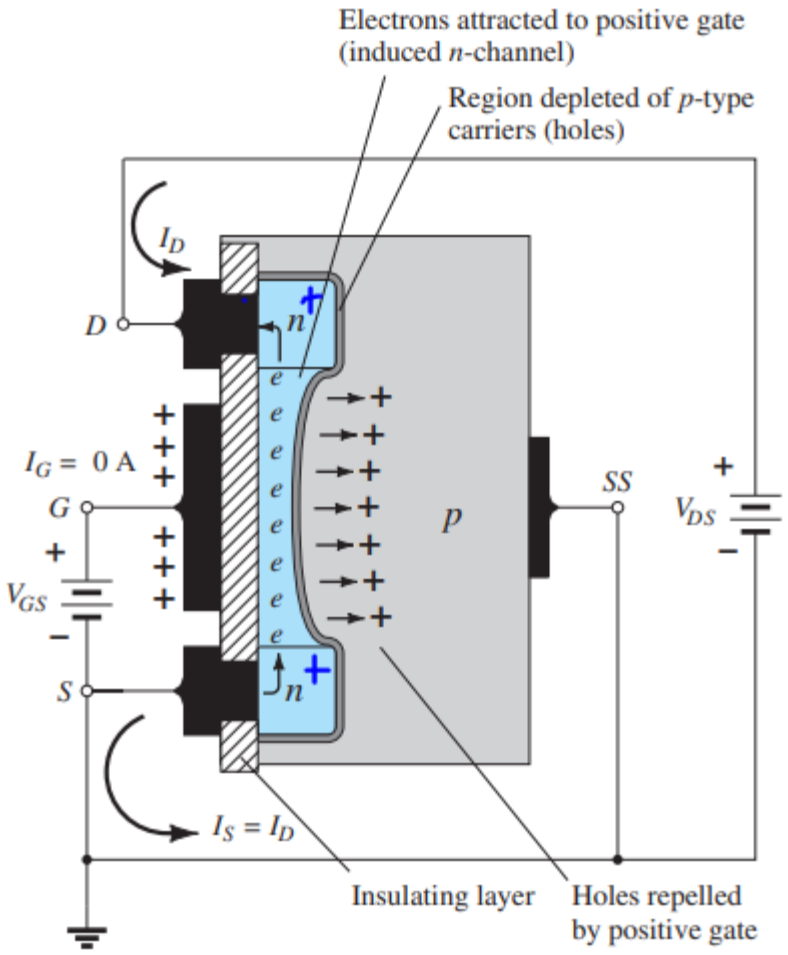


Fig. Channel formation in the n-channel enhancement-type MOSFET.

- ✓ Substrate always connected to the source.
- ✓ A voltage V_{GS} is connected between gate and source to make the gate +ve with respect to the source.
- ✓ The +ve gate voltage attracts electrons from the substrate to the region along the insulating layer (SiO_2) opposite to the gate.
- ✓ If the gate is made sufficiently +ve enough, electrons will be drawn in to that region to convert it to N-type material.
- ✓ Thus, an N-type channel will be formed b/n drain and source.

- ✓ **If positivity of the gate increases, the channel will widened and making it more conductive.**
- ✓ **Increasing the value of V_{GS}, enhances the conductivity of the channel and increases the flow of current from drain to source.**
- ✓ **Since electrons are induced in to the channel to convert it to N-type material, the MOSFET is called an induced N-channel Enhancement MOSFET.**
- ✓ **N-channel exist only when V_{GS} is positive.**
- ✓ **The induced n-channel does not become sufficiently conductive to allow drain current to flow until V_{GS} reaches certain threshold voltage V_T. (1V to 3V)**

✓ For values of V_{GS} less than the threshold level the drain current of an enhancement type MOSFET IS 0mA.

✓ $I_D = k(V_{GS} - V_T)^2$

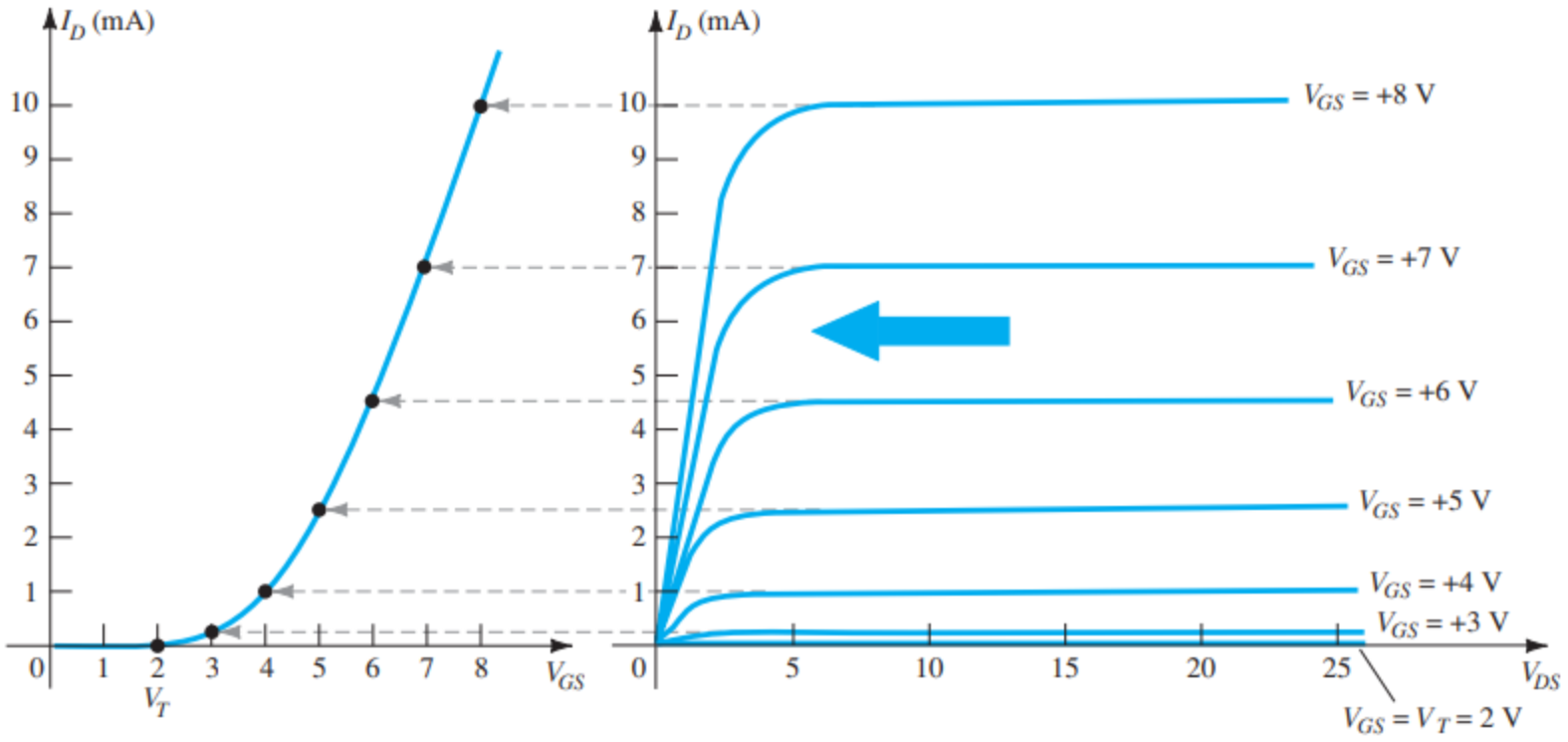


Fig. Sketching the transfer characteristics for an n-channel enhancement-type MOSFET from the drain characteristics.

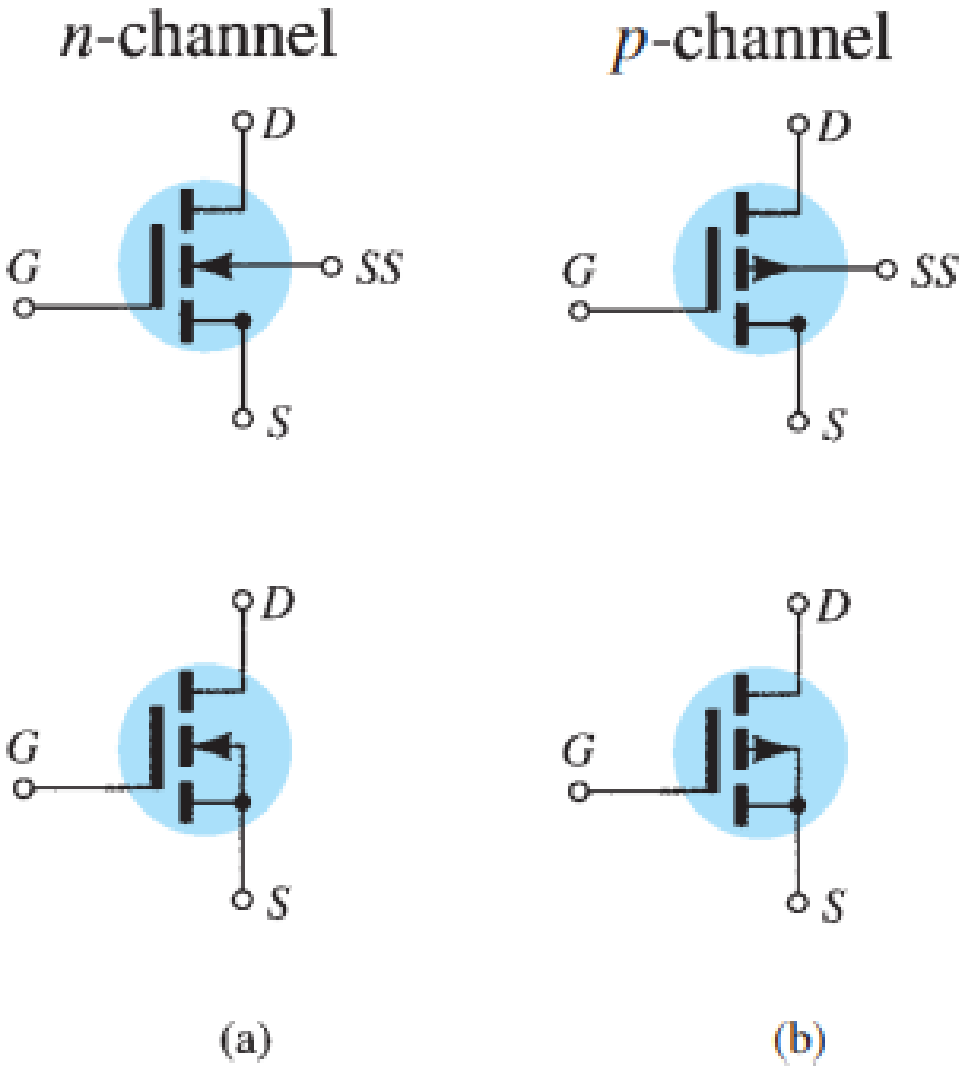


Fig. Symbols for: (a) n-channel enhancement-type MOSFETs and (b) p-channel enhancement type MOSFETs.